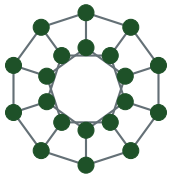
MID-SEMESTER STUDY SERIES GRAPH THEORY DAY 9

CS-5136: Graph Theory

Hamiltonian Graphs, Part 1 — Rebuilt From First Principles

pages 161–188 · every theorem built up
from a warm-up example, with an honest check
of the converse wherever a theorem is one-directional



Built from: A. Bhattacharjee, "CS-5136: Graph Theory", NIT Silchar
this is a from-scratch rebuild of Day 9, at a slower and more careful pace
this section is large — Day 10 finishes it (pages 189–194)

A One-Line Reminder From Day 8

Some theorems say *"A if and only if B"* — these need **two** proofs, one for each direction ($A \Rightarrow B$ and $B \Rightarrow A$), and we'll always label them clearly. Others say just *"if A then B"* with no claimed converse — for those, we'll explicitly check whether $B \Rightarrow A$ happens to be true anyway (often it isn't, and we'll show a small counterexample rather than leave you wondering). Ore's Theorem, later in this document, is exactly this second kind — flagged clearly when we get there.

1 The Icosian Game

In 1857, Sir William Rowan Hamilton invented a puzzle sold as a wooden regular dodecahedron: twenty corners, each named after a city, and the challenge was to travel "Around the World" — visiting every city exactly once and returning home. In graph terms: the dodecahedron's 20 vertices and 30 edges form a graph, and the puzzle asks for a cycle through every vertex exactly once. This is the motivating example for the whole chapter.

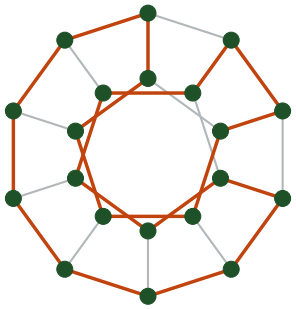
Definition 1.1 (Hamiltonian cycle / path)

A cycle in G that includes *every* vertex of G is a **Hamiltonian cycle**; G possessing one is a **Hamiltonian graph**. A **Hamiltonian path** is a walk containing every vertex exactly once (no requirement to return to the start).

Three quick facts — all one-directional and all easy to see why

A Hamiltonian cycle is strictly stronger than a Hamiltonian path: delete any one edge from a cycle and a path remains, so *cycle⇒path* — but not the reverse (a plain path graph P_4 has a Hamiltonian path trivially, itself, but has no cycle at all to speak of, so it certainly has no Hamiltonian *cycle*). Every Hamiltonian graph is connected (a cycle through every vertex visits them all, so they're mutually reachable) — but plenty of connected graphs aren't Hamiltonian, so this doesn't reverse either. And no single vertex removal can disconnect a Hamiltonian graph (the cycle minus one vertex is still a single path touching everyone else) — again, the reverse is false: staying connected after removing any one vertex doesn't guarantee a Hamiltonian cycle exists (we'll see graphs like this later in this very chapter).

Example — The dodecahedron graph solved

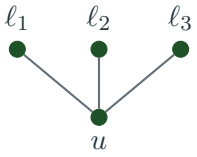


Grey: all 30 edges of the dodecahedron graph (outer 10-cycle, inner two-pentagon "star", and 10 connecting spokes — 3-regular throughout). Rust: one verified Hamiltonian cycle, visiting all 20 vertices and returning to the start — Hamilton's puzzle, solved.

2 A Necessary Condition

2.1 A tiny warm-up first

Before any formula, look at this small graph: a hub vertex u with three separate "arms", each a single leaf hanging off u (so u has degree 3, each leaf has degree 1).



Could this have a Hamiltonian cycle? A cycle needs to *enter and leave* every vertex it visits — but each leaf ℓ_i has only **one** edge, so a cycle passing through ℓ_i would need to use that same single edge twice (once to arrive, once to leave) — not allowed. So this graph obviously can't be Hamiltonian. The lemma below turns this "obvious" small observation into a general, useful tool.

Lemma 2.1

If G is Hamiltonian, then for every nonempty proper subset S of $V(G)$: $\kappa(G - S) \leq |S|$ (removing S leaves at most $|S|$ components).

Proof. Let C be a Hamiltonian cycle of G, and suppose $G - S$ has components $G_1, \ldots, G_k$. Walk around C. Every time C visits a component G_i for the *last* time before eventually leaving it for good, the very next vertex on C cannot be in G_i again (that visit was the last one) and can't be in a different component G_j directly either (there's no edge between different components of $G - S$ by definition) — so that next vertex must be in S. This happens (at least) once for each of the k components, at k different points along C (different components' "last visits" happen at different steps, since C visits only one vertex at a time) — giving k distinct vertices of S used this way. So $|S| \geq k = \kappa(G - S)$. ■

Is this an "if and only if"? No — and here's the corollary that makes it useful anyway

Lemma 2.1 only claims one direction (Hamiltonian ⇒ the inequality). The converse — *"the inequality holding for every S guarantees Hamiltonian"* — is not claimed, and isn't true in general (there are non-Hamiltonian graphs that still satisfy $\kappa(G - S) \leq |S|$ for every S; Ore's theorem later in this chapter is what it actually takes to *guarantee* Hamiltonicity). What Lemma 2.1 *is* perfect for is its own logical contrapositive, which is automatically just as true:

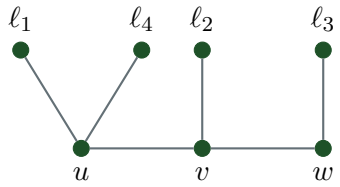
Corollary 2.2 (the contrapositive — logically the same fact restated)

If $\kappa(G - S) > |S|$ for *some* nonempty proper subset S, then G is **not** Hamiltonian.

(A contrapositive — "not $Y \Rightarrow$ not X" from "$X \Rightarrow Y$" — is not a second, separate direction needing its own proof; it's automatically true the moment the original statement is, by pure logic, regardless of subject matter. If it's raining ⇒ the ground is wet, then automatically: ground *not* wet ⇒ it did *not* rain.)

Example — A simple graph that fails the test

Take $S = \{u, v, w\}$, plus four more vertices $\ell_1, \ell_2, \ell_3, \ell_4$, each joined to *exactly one* of u, v, w (say ℓ_1, ℓ_4 to u; ℓ_2 to v; ℓ_3 to w), and no edges among the ℓ_i themselves.



Removing $S = \{u, v, w\}$ leaves $\ell_1, \ell_2, \ell_3, \ell_4$ with *no* edges between any of them — 4 isolated vertices, i.e. 4 separate components. $\kappa(G - S) = 4 > 3 = |S|$, so by Corollary 2.2, G is **not** Hamiltonian — confirmed instantly, with no need to search for (and fail to find) a cycle.

3 Ore's Theorem

3.1 A warm-up before the general statement

Take K_4 minus one edge (vertices a, b, c, d; every pair adjacent *except* a, b). Is it Hamiltonian? Just try: a, c, b, d, a — check each step is a real edge (ac✓, cb✓, bd✓, da✓) — yes, a genuine Hamiltonian cycle, found by hand in seconds. Notice: the only nonadjacent pair is (a, b), and $d(a) + d(b) = 2 + 2 = 4$, while $n = 4$ — so the degree sum *equals* n exactly. Ore's theorem says this was no accident: whenever nonadjacent degree sums are big enough (at least n), you never even have to search — a Hamiltonian cycle is guaranteed.

Theorem 3.1 (Ore's Theorem 1960)

Let G have order $n \geq 3$. If $d(u) + d(v) \geq n$ for *every* pair of nonadjacent vertices u, v, then G is Hamiltonian.

Flagging this clearly: only one direction is claimed

This theorem is *not* an "if and only if". It says *big-degree-sums* $\Rightarrow$ *Hamiltonian*, full stop. Once we've proved that direction, we'll explicitly check — rather than just assume — whether the reverse (*Hamiltonian* $\Rightarrow$ *big degree sums*) also happens to hold. Spoiler: it doesn't, and we'll show you the exact graph that breaks it.

Proof. Suppose, for contradiction, some graph G satisfies the hypothesis but is *not* Hamiltonian.

Step 1 — push it to the edge of failure. Keep adding edges to G (never removing any) for as long as the graph stays non-Hamiltonian, until no more edges can be added without creating a Hamiltonian cycle. Call the result H. Since we only ever *added* edges, every vertex's degree in H is at least what it was in G — so H *still* satisfies $d(u)+d(v) \geq n$ for all its nonadjacent pairs (degrees only went up). H itself is not Hamiltonian, but adding *any* single edge to it creates a Hamiltonian cycle.

Step 2 — find a Hamiltonian path. H isn't complete (else it would trivially be Hamiltonian for $n \geq 3$), so pick a nonadjacent pair x, y in H. Adding edge xy creates a Hamiltonian cycle in $H + xy$; that cycle must use the new edge xy (else it would already have existed in H, contradicting H non-Hamiltonian) — so removing just that edge from the cycle leaves a **Hamiltonian path** in H itself, from x to y. Label its vertices in order: $x = x_1, x_2, \ldots, x_n = y$.

Step 3 — the key counting trick. For each i from 2 to n: *if $x_1 x_i$ is an edge of H, then $x_{i-1} x_n$ is not.* Why: if both were edges, then

$$x_1,\ x_i,\ x_{i+1}, \ldots, x_n,\ x_{i-1},\ x_{i-2}, \ldots, x_1$$

would be a Hamiltonian cycle of H (follow the path forward from x_i to x_n, jump back to x_{i-1} using the assumed edge, then follow the path *backward* from x_{i-1} to x_1) — but H is non-Hamiltonian. Contradiction. So no such i can have both edges.

Step 4 — turn that into a degree bound. Every neighbour of x_1 is some x_i with $i \geq 2$; Step 3 says each such i forces x_{i-1} to be a *non*-neighbour of x_n. Different i's give different $i-1$'s, so this pairs up each of x_1's $d(x_1)$ neighbours with a *distinct* non-neighbour of x_n among $\{x_1, \ldots, x_{n-1}\}$ (a set of size $n-1$). Since x_n has $d(x_n)$ neighbours total, all lying in that same $(n-1)$-vertex set, the number of *non*-neighbours available there is exactly $(n-1) - d(x_n)$. So:

$$d(x_1) \ \leq\ (n-1) - d(x_n) \quad \Longrightarrow \quad d(x_1) + d(x_n) \ \leq\ n - 1.$$

Step 5 — contradiction. But $x_1 = x$ and $x_n = y$ are *nonadjacent* (that's how we picked them), so the theorem's hypothesis demands $d(x) + d(y) \geq n$ — directly contradicting Step 4's $d(x_1) + d(x_n) \leq n - 1$.

This contradiction means no such non-Hamiltonian G can exist. ■

Example — Checking the converse honestly: does it fail?

Take $G = C_5$ (the 5-cycle). It's obviously Hamiltonian — it *is* a cycle through all 5 vertices, trivially. Now check Ore's hypothesis: every nonadjacent pair in C_5 has degree sum $2 + 2 = 4$. Is $4 \geq n = 5$? **No.** So C_5 is a Hamiltonian graph that fails Ore's hypothesis completely — proving conclusively that *Hamiltonian does NOT imply big degree sums*. The converse of Ore's theorem is false, and now you've seen exactly why: Ore's condition is a comfortable, sufficient *shortcut* to Hamiltonicity, not a requirement every Hamiltonian graph must meet.

3.2 Closing a loop from Day 4: the closure connection

Theorem 3.2

A graph G is Hamiltonian **if and only if** its closure $C(G)$ (Day 4: repeatedly join nonadjacent pairs with degree sum $\geq n$) is Hamiltonian.

This one really is an "if and only if" — both directions below, labelled clearly.

Proof (Direction 1: G Hamiltonian $\Rightarrow$ $C(G)$ Hamiltonian). Closure only *adds* edges, never removes any — so a Hamiltonian cycle already present in G survives completely untouched in $C(G)$; it's still right there, still visiting every vertex, still using only edges that are still present. ■

Proof (Direction 2: $C(G)$ Hamiltonian $\Rightarrow$ G Hamiltonian — the harder direction). Suppose $C(G)$ is Hamiltonian. The closure is built one qualifying edge at a time: $G = C_0 \subset C_1 \subset \cdots \subset C_m = C(G)$, where $C_j = C_{j-1}$ plus one new edge $x_j y_j$ satisfying $d_{C_{j-1}}(x_j) + d_{C_{j-1}}(y_j) \geq n$ at the moment it was added. Suppose, for contradiction, $G = C_0$ is *not* Hamiltonian. Since C_m *is*, there's some first step j where non-Hamiltonian C_{j-1} becomes Hamiltonian C_j by adding just the one edge $x_j y_j$. That is exactly the situation Ore's proof analyzes (Steps 2–4 above, with $H = C_{j-1}$, $x = x_j$, $y = y_j$): it shows $d_{C_{j-1}}(x_j) + d_{C_{j-1}}(y_j) \leq n - 1$. But $x_j y_j$ was only *eligible* to be added to the closure because $d_{C_{j-1}}(x_j) + d_{C_{j-1}}(y_j) \geq n$ — contradiction. So no such first step exists, meaning $G = C_0$ must already have been Hamiltonian. ■

Why this matters

This is the whole point of building the closure at all: instead of checking Ore's exact degree-sum condition on G (which might fail), you can keep adding "free" edges (each individually justified by $\geq n$) until either you reach K_n (trivially Hamiltonian, so G was too) or you get stuck with a graph that's obviously simpler to analyze — all without ever risking a wrong answer, by Theorem 3.2, now proved in *both* directions.

4 Day 9 Cheat-Sheet

Recap: which theorems are "iff" and which aren't

- **One direction only (converse checked and found false):** Lemma 2.1 ($\kappa(G-S) \leq |S|$ is necessary, not sufficient); Ore's Theorem (C_5 shows Hamiltonian graphs can badly fail the degree-sum test).
- **Iff (both directions proved):** G Hamiltonian $\iff$ $C(G)$ Hamiltonian.
- Necessary condition: Hamiltonian $\Rightarrow \kappa(G-S) \leq |S|$ for every proper nonempty S. Contrapositive (same fact, restated): find one bad S to instantly rule out Hamiltonicity.
- Ore's theorem proof idea: push to a maximal non-Hamiltonian graph, find the Hamiltonian path forced by almost-Hamiltonicity, then count neighbours vs. non-neighbours to force a contradiction.
- Closure theorem reuses *exactly* Ore's counting argument, applied one added edge at a time.

5 Practice Questions

Set A — definitions and the necessary condition

Q1. Is the star graph $K_{1,4}$ (one hub, four leaves) Hamiltonian?

A. *No. Take $S = \{hub\}$: removing it leaves 4 isolated leaves — $\kappa(G-S) = 4 > 1 = |S|$, so by Corollary 2.2, not Hamiltonian.*

Q2. True or false: "if $\kappa(G-S) \leq |S|$ holds for every possible S, then G must be Hamiltonian."

A. *False — this is exactly the converse of Lemma 2.1, which was never claimed and isn't guaranteed. Passing this test for every S is necessary but not sufficient.*

Set B — Ore's theorem and closure

Q3. G has order 5 with degree sequence $4, 4, 3, 3, 2$. Can Ore's theorem be applied directly?

A. *Only if* every *nonadjacent pair sums to ≥ 5. The only vertex that could be a problem is the degree-2 one: if it's nonadjacent to another vertex of degree 2 or less, the sum could fall below 5. Without knowing the exact adjacency (not just the degree sequence), we can't confirm Ore's hypothesis from the sequence alone.*

Q4. If $C(G) = K_n$, what can you conclude about G?

A. *K_n is Hamiltonian for $n \geq 3$. By Theorem 3.2's Direction 2, G itself must then be Hamiltonian too — even though G might have very few edges before closure fills the rest in.*

Q5. Find another graph, besides C_5, that is Hamiltonian but fails Ore's hypothesis.

A. *Any cycle C_n for $n \geq 5$: every vertex has degree 2, so any nonadjacent pair sums to 4, while $n \geq 5$ — the hypothesis fails, yet C_n is trivially Hamiltonian (it's a cycle by construction). The bigger n gets, the more dramatically Ore's test fails while Hamiltonicity remains obviously true.*

Just say "Day 10" to finish this chapter (pages 189 onward: a degree-counting theorem, and counting Hamiltonian cycles in K_n).